HDOC DAY 55

PROBLEM

Write a C program that reads the size and elements of a variable-length integer array, arr[], and an integer k such that 0 < k ≤ size. Pass the array to a function named window_maximum() using its address. Use pointers to pass the array between the calling function and window_maximum(). The function window_maximum() must:

1. Consider every contiguous subarray of size k as a sliding window.
2. Find the maximum element in each window while moving from left to right.
3. Display the maximum elements in the order of their corresponding windows.

Separate the maximum elements by spaces.

4. Display -1 if k is invalid.
5. Use a nested-loop approach to solve the problem.

CODE

```
#include <stdio.h>

#include <stdlib.h>


void window_maximum(int *arr, int size, int k)

{

  if (k <= 0 || k > size)

  {

    printf("-1");

    return;

  }


  for (int i = 0; i <= size - k; i++)

  {

    int max = *(arr + i);
```

```
        for (int j = i + 1; j < i + k; j++)
        {
            if (*(arr + j) > max)
            {
                max = *(arr + j);
            }
        }

        printf("%d", max);

        if (i < size - k)
            printf(" ");
    }
}

int main()
{
    int size, k;

    scanf("%d", &size);

    if (size <= 0)
    {
        printf("-1");
        return 0;
    }
```

```
    int *arr = (int *)malloc(size * sizeof(int));

    if (arr == NULL)
        return 1;

    for (int i = 0; i < size; i++)
    {
        scanf("%d", arr + i);
    }

    scanf("%d", &k);

    window_maximum(arr, size, k);

    free(arr);

    return 0;
}
```